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EEE 323 — Past Questions & Materials

by Renance

Course: EEE 323

Institution: Federal University of Petroleum Resources, Effurun

Papers in this pack: 2

Levels: 300

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Paper 1 of 2 — EEE 323 ELECTROMAGNETIC FIELDS AND THEORY II

Level 300 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

COLLEGE OF ENGINEERING AND TECHNOLOGY

Department of Electrical and Electronics Engineering

EXAMINATION: Second Semester COURSE CODE: EEF. 323

SESSION: 2023/2024 COURSE TITLE: Electromagnetic Fields and Theory #1

DEGREE: B.Eng TIME ALLOWED: 3 Hours

DATE: October 11, 2024

Question One

(a) Briefly discuss the following terms as they relate to a transmission line used to propagate electromagnetic waves: (i) propagation constant (ii) characteristic impedance (iii) standing wave ratio (iv) reflection coefficient

(b) A lossless transmission line is 200m long and operates at a frequency of 600 MHz. The line parameters are $L = 0.25 \mu\text{H/m}$, $C = 100 \text{ pF/m}$. Find i. the characteristic impedance ii. the phase constant iii. the phase velocity.

(c) A transmission line of 2Km length is terminated in load equal to its characteristic impedance. It is excited by a generator with voltage of 300V. If the propagation constant on the line is $0.7 + j0.3$ per Km, determine the amplitude of the voltage at the middle of the line.

/ Question Two

(a) Briefly discuss the following terms associated with a communication antenna: i. Antenna efficiency (ii) directive gain (iii) antenna power gain. (iv) antenna polarization. (v) Radiation resistance of an antenna.

(b) In the construction of an outdoor television antenna, hollow tubular conductors are usually used instead of solid conductors. Also, silver plating is often used to reduce material cost of the waveguide components. Briefly explain the reason behind this electrical engineering professional practice

Transmission coefficient (U)

Transmit antenna gain

line loss (α_{dB}) BRP

Resistor

CC

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Transmission $p < s > - |$

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Earth's Surface

Figure |

Given the free-space radio transmission system shown in fig 1 with the following transmission characteristics:

Power ovi (Pour)= 60dBm. transmission line loss, $L_y = 4\text{dB}$, free-space loss, $L_p = 50\text{dB}$. Determine

1. the antenna input power (P_{in}),
 - ii. ii. radiated power ((P_{raa}),
 - iii. lil, Effective isotropic radiated power (EIRP)
 - iv. iv. Receive power density (P)
- for a transmit antenna with directivity of unity ($D, = 10$) and efficiency, $\eta = 50\%$.

- v. the power received in watts and dB when the operating frequency is 1 GHz and the directivity of the receive antenna is unity.

Number Three (14 marks)

- (a) Discuss briefly what you understand by impedance matching in transmission lines?
- (b) Discuss three methods used for impedance matching in a transmission line.
- (c) Due to losses, the waveguides are usually preferred to open wires and coaxial cables at very high frequencies. Briefly discuss the three types of losses associated with these open wires and coaxial cables at very high frequencies.
- (d) A microwave engineer is provided with a rectangular waveguide measuring $4 \times 2 \text{ cm}$. The following options of modes of transmission are available for him to choose: $TE_{9,1}$, $TE_{9,2}$, $TE_{1,1}$. Find all the possible modes that can be chosen by the engineer to ensure propagation through the guide when the operating frequency is 5 GHz.

; Number Four (14 marks)

- (a) Explain (i) ground wave, (ii) space wave and (iii) the sky wave modes of propagation. State the area of application of each mode.
- (b) Differentiate between service range and skip distance in sky mode of propagation.
- (c) Assuming that reflection takes place at a height of 300 Km and that the maximum density in the ionosphere corresponds to 0.75 refractive index at 12 MHz, what will be the range (skip distance) assuming the earth is flat for which the maximum usable frequency is 14 MHz.

Number Five (14 marks)

(a) Consider an isotope located at point O, the center of sphere, fed with power P watts in fig 2. Show that the electric field developed by this isotope at point Q, a distance of r meters from the center is given by $E = \frac{\sqrt{30P}}{r^2}$

Figure 2 :

(b) Find the velocity of plane wave in a lossless medium having a relative permittivity of 20 and relative permeability of 5

(c) If the electric field strength of plane wave is 12 V/m, What is the strength of the magnetic field H in free space?

(d) Find the electric field strength at 1 Km away from an isotropic station radiating 2 KW of power

Question Six

(a) What do you understand by (i) Transverse electric, TE waves (ii) Transverse magnetic, TM (iii) Transverse electromagnetic TEM wave in wave propagation. (6 marks)

(b) A television transmitting antenna mounted at a height of 100m radiates 10Kw of power equally in all directions of azimuth at a frequency of 50 MHz, Calculate (i) the maximum line-of-sight range (ii) the field strength at the receiving antenna mounted at a height of 9m at a distance of 10 Km. (6 marks)

Question Seven .

(a) State the four Maxwell's equations in integral form in time varying field and explain the law or phenomenon each represents in electromagnetics.

(b) In the air space within a large power transformer, the magnetic flux density is given by $H = 10 \cos(377t + 1.60 \times 10^3 z)$ ay A, Determine (i) the magnetic flux density, B. (ii) the amplitude of the displacement current density. Assume the conduction current density (J) in the air space is zero.

(c) Calculate the power density reaching the Moon's surface from a 20 MW pulse transmitter located on the earth. The antenna gain is 65dB. Take the distance between the moon and the earth as 400,000Km. Assume the transmission medium is lossless,

Paper 2 of 2 — EEE 323 Electromagnetic Field and Theory

Level 300 • Second Semester • PastQuestion

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Question Seven i a? tiiligsas

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

COLLEGE OF TECHNOLOGY

Department of Electrical and Electronic Engineering

EX4M!NATION: 2nd Semester COURSE CODE: EEE 323

SESSION: 2

018/2019 COURSE TITLE: Electromagnetic Field Theory II

DEGREE: 8. Eng.

DATE: September 28, 2019

INSTRUCTION: Answer any five questions

Constants: $\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$, $\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$, $c = 3 \times 10^8 \text{ m/s}$

Number One (14 marks)

(a) I:xpain the following terms associated with transmission lines (i) propagation

constant (ii) characteristic impedance (iii) standing wave ratio (iv) reflection coefficient

(b) Explain skin effect in transmission lines. State two arcas of appiication of this effect.

(c) A line leading to a broadcast transmitting antenna operates ata frequency of 2 Mikiz MHz. The length the line 100m and has a characteristic impedance of 609. What is the

input impedance if the remote end is (i) open circuited (ii) short circuited,

Number Two (14 marks)

- (a) What is impedance matching in a transmission line?
- (b) Discuss three methods used for impedance matching in a transmission line.
- (c) Briefly discuss the three types of losses associated with these transmission lines at very high frequencies.
- (d) A half wavelength antenna tuned to 20 MHz is equivalent to a non-inductive resistance of 100 Ω between its feed points. It is fed by a parallel wire feeder with a characteristic impedance of 500 Ω . Design a suitable $\lambda/4$ wave transformer to provide

matching between them i.e. determine the length and characteristic impedance of the $\lambda/4$ wave transformer

Number three (14 marks)

- (a) Explain (i) ground wave, (ii)

State the area of application of each mode,

- (b) Differentiate between service range and skip distance in sky mode of propagation.

(by Two aircrafts are flying at altitudes of 3000 m and 5000 m respectively, What's the distance along the surface of the earth over which they can have

communication? Take the actual curvature of the earth

space wave and (iii) the sky wave modes of propagation.

maximum possible

effective point to point microwave communication

into consideration.

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Question Four (14 marks)

- (a) Write the set of four Maxwell's equations:

(i) Set of four Maxwell's equations both in differential and integral forms in a time varying field i.e. "static" . . .

(ii) Magnetic field. Briefly describe the meaning of each of these equations with particular reference to the laws/theorems of electromagnetics they belong to.

(iii) Determine the expressions for electric field density D , magnetic flux density B , and the magnetic field intensity H in free space given that the electric field intensity, E , is $\sin(\omega t - \beta z)$ a.u. —

by Dz - Sot

- $\epsilon_0 \nabla \cdot E = \rho$ -

Number Five (14 marks)

- (a) Define the following (i) Antenna (ii) Antenna efficiency (iii) Directive gain of an antenna (iv) The effective area of an antenna (v) Isotropic antenna

(b) i.. What is radiation pattern of an antenna? ——— C

ii, State the three principle field regions surrounding an antenna

(c) The radiation resistance of an antenna is 8Ω . The efficiency factor is 0.9. Calculate the loss resistance R_{loss} if the magnitude of the current, $I = 4A$. Calculate the power radiated and the ohmic loss.

Two Marks (14 marks)

a) i. What is polarization of an antenna?

ii. Draw the three types of polarization

(iii) Draw the three types of polarization

(b) Describe a non-resonant antenna and state its demerits

(c) With the aid of suitable diagram(s), discuss with sufficient details, the Half wave dipole antenna, >

> Question Seven (14 marks)

4) What is a distortionless transmission line?

(b) Under what condition is a transmission line said to be distortionless?

(c) A rectangular waveguide has the following dimensions: $a = 10cm$, $b = 5cm$. The filler is required to pass electromagnetic wave with TE_{10} configuration. Calculate the waveguide's (i) cutoff frequency (ii) cutoff wavelength (iii) characteristic impedance.

Take the operating frequency to be 5 MHz.

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SESSION: 2020/2021

COURSE TITLE: Electromagnetic Field Theory II

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DEGREE: B. Eng. ' "

8 ARS Bed TIME ALLOWED: 3 Hours

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DATE: December 1, 2022

INSTRUCTION: Answer any five questions:

8.85×10^{-12} F/m, $c = 3 \times 10^8$ m/s

Question One.

(a) i. What is an antenna? ii. State the two basic functions of an antenna. iii, Define the following antenna parameters: antenna efficiency, antenna power gain, effective area of an antenna, and radiation resistance [4 marks] ' "

(b) What is an 'isotropic antenna? [2 marks]

(c) Consider an isotropic antenna at point O in figure | fed with power P_{in} in spherical surface S with a radius r . Show that the electric field intensity at point Q in the sphere is given by $E = \sqrt{30P_{in}/r}$ V/m. [4 marks] Nas QF ok

(d) An antenna has a loss resistance of 25.0, power gain of 30 and directivity of 42. a Ke *

Constants: $\mu_0 = 4 \times 10^{-7} \text{ H/m}$, $\epsilon_0 =$

Se

E 4%

Calculate its radiation resistance. [4 marks] oY na)

$\tau_m * \omega =$

SA Question Two ny

(a) List the primary and secondary constants of a transmission line [2 marks] Y

(b) Write the expressions and explain the following terms associated with transmission' lines (i) propagation constant (ii) characteristic impedance (iii) standing wave ratio

(iv) reflection coefficient. [6 marks]

(c) A uniform transmission line has: $R = 500/\text{km}$, $L = 1\text{mH/km}$, $G = 1\text{U/km}$, $C = 0.1\text{uF/km}$ at $\omega = 6000\text{radians/second}$. Calculate (i) the characteristic impedance of the line, Z_0 , (ii) the propagation constant, γ . [6 marks]

fo estion Three

(a) Explain (i) ground wave, (ii) space wave and (iii) the sky wave modes of propagation. State the area of application of each mode. [6 marks]

4(b) Differentiate between service range and skip distance in sky mode of propagation, [2 marks] ; :

(C) Two aircrafts are flying at altitudes of 4000 m and 7000 m respectively. What is the maximum possible distance along the surface of the earth over which they can have effective point to point microwave communication? Take the actual curvature of the earth into consideration. [6 marks]

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(a) What do understand by (i) Tr:

(iii) Transverse electromagnetic TEM wave

(b) Explain skin effect in transmission line

Ne i oe antenna mounted at a height of 120m radiates 25Kw of power equally

a caret eee ala frequency of 50 MHz. Calculate (i) the maximum line-of-

lg © (1) the field strength at the receivin antenna mount i

distance of 12 Km, [6 marks] ° "eave cei of Tom ata

ansverse electric, T waves (ii) Transverse magnetic, Ty

€ in wave propagation. [6 marks]

S [2 marks] -

Question Five

(a) Write the set of four Maxwell's

Briefly describe the meaning of

laws/theorems of electromagnetism

(b) Find the amplitude of displacement current density inside a typical metallic conductor having a conductivity $\sigma = 5 \times 10^7 \text{ S/m}$, $\epsilon = 1$, $f = 1 \text{ kHz}$ for current density $J = 10 \sin(\omega t - 4442) \text{ A/m}$? [6 marks]

Write Maxwell's equation in integral form in a time varying field.

Relate each of these equations with particular reference to the

physics they belong to. [8 marks]

Question Six

(a) What is impedance matching in a transmission line? [3 marks]

(b) Discuss two methods used for impedance matching in a transmission line, (2 marks)

(c) Briefly discuss the three types of losses associated with open transmission wires and coaxial cable at very high frequencies. [3 marks]

(d) i. What do you understand by dominant mode of propagation in a waveguide? [4 Marks]

ii. Refer to figure 6. Determine the longest cut-off wavelength for this waveguide operating in TE_{10} mode. [4 marks]

Question Seven

(a) Define critical frequency and maximum usable frequency as they relate to sky wave propagation. [5 marks]

(b) Calculate the value of the critical frequency at which an electromagnetic wave must propagate through the D-Region with an index of refraction of 0.5 and an electron density of $3.25 \times 10^4 \text{ electrons/m}^3$? [5 marks]

(c) Find the velocity of plane wave in a lossless medium having relative permittivity $\epsilon_r = 8$ and relative permeability $\mu_r = 1$. [4 marks]

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SESSION: 2017/2018 COURSE TITLE: Electromagnetic Field Theory I

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'GREE: B. Eng. TIME ALLOWED: 2 Hours

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DATE: October 6, 2018

INSTRUCTION: Answer any five questions

Constants: $\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$, $\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$, $c = 3 \times 10^8 \text{ m/s}$

Number One (14 marks)

(a) Explain the following terms as they relate to electromagnetic waves propagating through a transmission line (i) propagation constant (ii) characteristic impedance

(1H) standing wave ratio

(5) A uniform transmission line has: $R = 400/\text{km}$, $L = 1\text{mH}/\text{km}$, $G = 1\text{G}/\text{km}$, $C = 0.1\text{pF}/\text{km}$ at $\omega = 5000\text{radians/second}$. Calculate (i) the characteristic impedance of the line, Z_0 . (ii) the propagation constant, γ .

Number Two (14 marks)

(a) Due to losses, the waveguides are usually preferred to open wires and coaxial cables at very high frequencies. Briefly discuss the three types of losses associated with these transmission lines at these frequencies.

(b) A rectangular waveguide has the following dimensions: $a = 10\text{cm}$, $b = 5\text{cm}$. The filter is required to pass electromagnetic wave with TE_{10} configuration. Calculate the waveguide (i) cutoff frequency (ii) cutoff wavelength (iii) characteristic impedance

Number Three (14 Marks)

(a) Explain (i) ground wave, (ii) space wave and (iii) the sky wave modes of propagation. State the area of application of each mode.

(b) A television transmitting antenna mounted at height of 140m radiates 20Kw of power equally in all directions at a frequency of 60MHz

(i) Calculate the maximum line-of-sight taking into consideration the effective radius of the earth: (ii) the electric field strength of the receiving antenna mounted at a distance of 16Km and (iii) the distance at which the field strength reduces to 2mV/m mounted on a height of 20m .

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distance of

The receiver is

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Number Four (14 marks)

(a) Define critical frequency of propagation,

(b) Assuming that reflection coefficient in the ionosphere is 1, calculate the range (skip distance) for a wave of frequency 16MHz ,

and maximum usable frequency as they relate to sky wave propagation. Assume that the maximum electron density in the ionosphere corresponds to 0.75 refractive index at 12MHz , what will be the maximum usable frequency if the earth is flat for which the maximum usable

Number Five (14 marks)

(a) Explain what you understand by impedance

; matching in a transmission line

(b) Discuss two methods

used for impedance matching in a transmission line.

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Q. (c) A lossless transmission line with $Z_0 = 500$ is 30m long and operates at 2 MHz. The line is terminated with a load $Z_L = (80 + j40)$. If the wave velocity, $V_p = 0.6C$, where C is the speed of light on the line, find

(i) the reflection coefficient Γ (ii) the standing wave ratio S (iii) the input impedance

Number Six (14 marks)

(a) What is an antenna? (2 marks)

(b) Determine the gain of a 3m diameter parabolic reflector at 4 GHz assuming 55% aperture efficiency. Express your answer in decibel, (5 marks)

(c) Calculate the minimum peak transmitter power needed in a pulsed radar required to a target of 15m² echoing area at a range of 100km given the following system parameters: operating frequency 1.5 GHz, receiver sensitivity -120 dBm (dB relative to 1mW), antenna gain 40 dB and atmospheric attenuation 0.009 dB/km. (7 marks)

Number Seven (14 Marks)

(a) Define the following antenna characteristics

i. Power gain, G (2 marks)

ii. Directivity, D (1 % marks)

iii. Antenna Efficiency ; (2 marks)

iv. Effective Aperture, A_e (1% marks)

Compute the peak power density at 10km of an antenna with an input power of 3 W and a gain of 15dB. Also determine the electric field intensity and the magnetic field strength. (7 marks)

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